

Salinity stress on hydroponics

Overview

Applying salinity stress at different time periods to Arabidopsis on hydroponics to evaluate whether our protectant are effective at rescuing Arabidopsis.

A. Materials

- a. $\frac{1}{2}$ MS medium powder
- b. MES buffer
- c. Distilled water
- d. Sucrose
- e. Agar powder
- f. NaCl
- g. 2000 ml beaker
- h. 500 ml flask
- i. pH meter
- j. Hot plate
- k. Magnetic stirring rod
- l. Aluminum foil
- m. Tweezers
- n. Eppendorf microtube
- o. Scissors
- p. Leather punch
- q. Tube rack

B. Procedure

Prepare eppendorf lids

1. Use scissors to cut off the lid of the eppendorf microtube.
2. Use leather punch to punch a hole on the eppendorf lid with a diameter of 3 mm.
3. Put the lids into a small beaker.
4. Wrap the beaker with aluminum foil and stick a piece of indicator tape on top.
5. Autoclave for 20 minutes at 127°C.
6. Wait for the pressure to reach 0 before opening the autoclave.

Prepare 1% MS agar for seed vernalization

1. The following protocol is for 250 ml of 1% agar.

$\frac{1}{2}$ MS powder (w/o sucrose)	0.575g
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MES buffer	0.125 g
Distilled water	250 ml
Sucrose (1%)	2.5 g
Agar powder (1%)	2.5 g

2. Add the ½ MS powder, MES buffer, distilled water, and sucrose into a 500 ml flask.
3. Use a hot plate to stir (with a magnetic stirring rod) until no sediments are left in the solution.
4. Use the pH probe and measure the pH of the solution, make sure the pH is 5.7-5.8. If the pH is too low, add 0.1M KOH slowly to increase it.
5. Add agar powder into the solution.
6. Wrap the neck of the flask with aluminum foil and stick a piece of indicator tape on top.
7. Autoclave for 20 minutes at 127°C.
8. Wait for the pressure to reach 0 before opening the autoclave.

Plant arabidopsis seeds in eppendorf lids

1. Prepare 48 eppendorf microtube lids.
2. Add 250 µL of agar into each lid.
3. Wait until it solidifies.
4. Soak a tube rack in bleach for 10 minutes and rinse it with ddH₂O until there is no smell of bleach on the tube rack.
5. Add 2ml of 0.1x MS medium into 48 wells in the tube rack.
6. Put the eppendorf lids onto the wells using tweezers.
7. Use an autoclaved toothpick to pick up one seed and plant arabidopsis seed onto the agar from the hole of the lid.
8. Wrap the tube rack with aluminum foil.
9. Place the tube rack into the refrigerator at 4°C for 4 days to stratify.
10. Put the tube rack into the growth chamber after 4 days of stratification.

Prepare 3M NaCl in ½ MS medium

1. Prepare the following materials:

½ MS powder (without sucrose)	0.92 g
MES buffer	0.2 g
Distilled water	400 mL

NaCl	70.128 g
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2. Add $\frac{1}{2}$ MS powder, MES buffer, and ddH₂O into a 500mL flask
3. Use the hot plate to stir (with a stirring rod) until no sediments are left in the solution
4. Determine pH of the solution and adjust the pH using a pH probe to around 5.7-5.8 using 1M KOH
5. Add 70.128 g of NaCl in batches slowly into the flask. Use the stirring rod to dissolve all NaCl into the medium.
6. Wrap the neck with aluminum foil and place the indicator tape on top
7. Autoclave at 127°C for 20 minutes
8. Wait for the pressure to reach 0 before opening the autoclave

Salinity stress

Hydroponics Salinity Testing

